

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

***Artificial Intelligence
and Society***



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 2.1: Neural Networks

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Research group

Complex Social & Computational Systems (CS)²

IDEa_lab

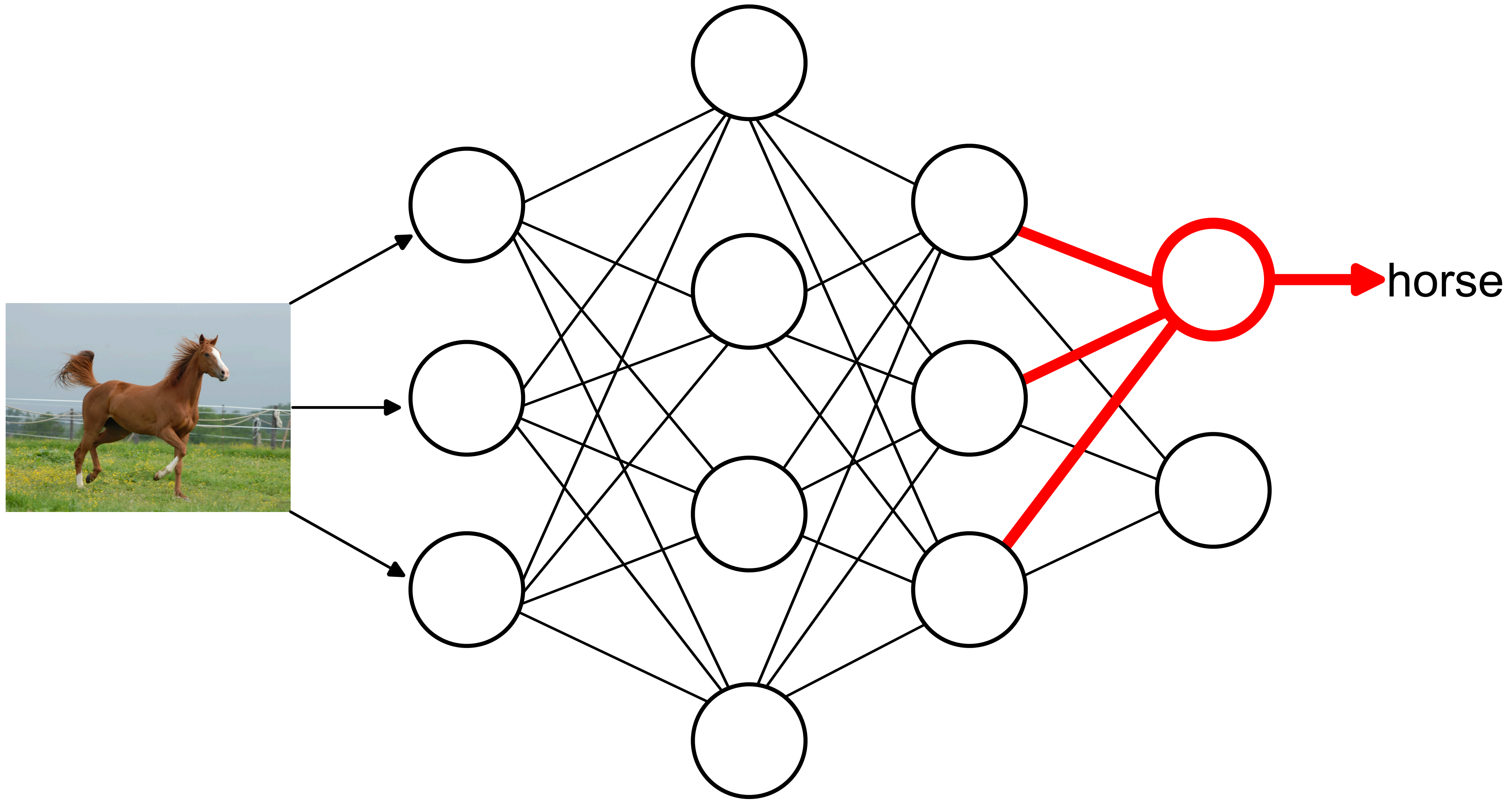
Das Interdisziplinäre Digitale
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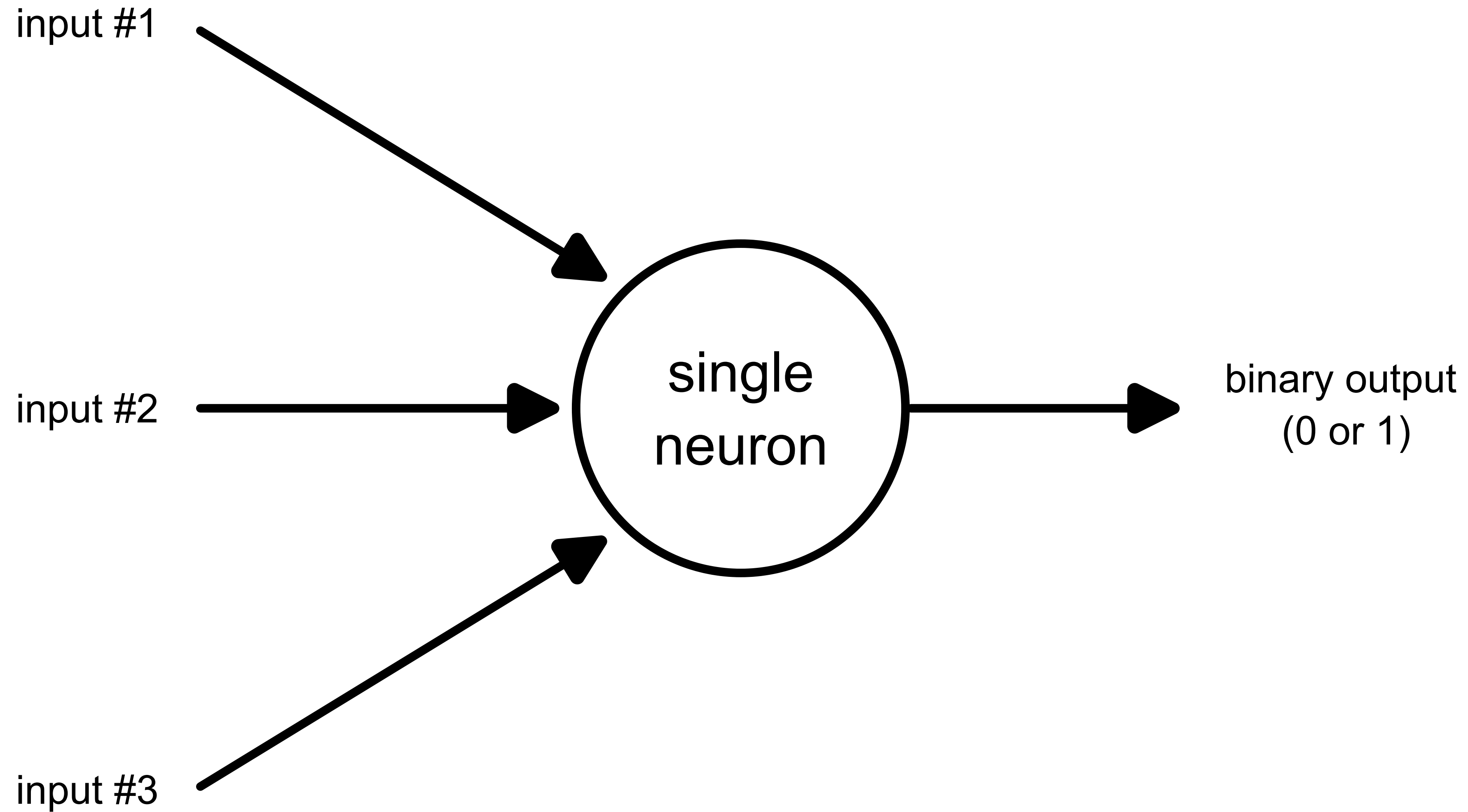
Supervised Learning Algorithms

- A machine learning algorithm for classification:
 - Takes the **attributes** x_i of an unknown observation as input
 - Predicts the **class** y_i of the observation (output).
 - **Learns** from training data to improve its **performance**.
- Last lecture: various algorithms for classification:
 - Naive Bayes
 - k -Nearest Neighbors
 - Decision Trees
- Now: neural networks as a powerful and flexible *architecture* for machine learning algorithms.

Neural Networks

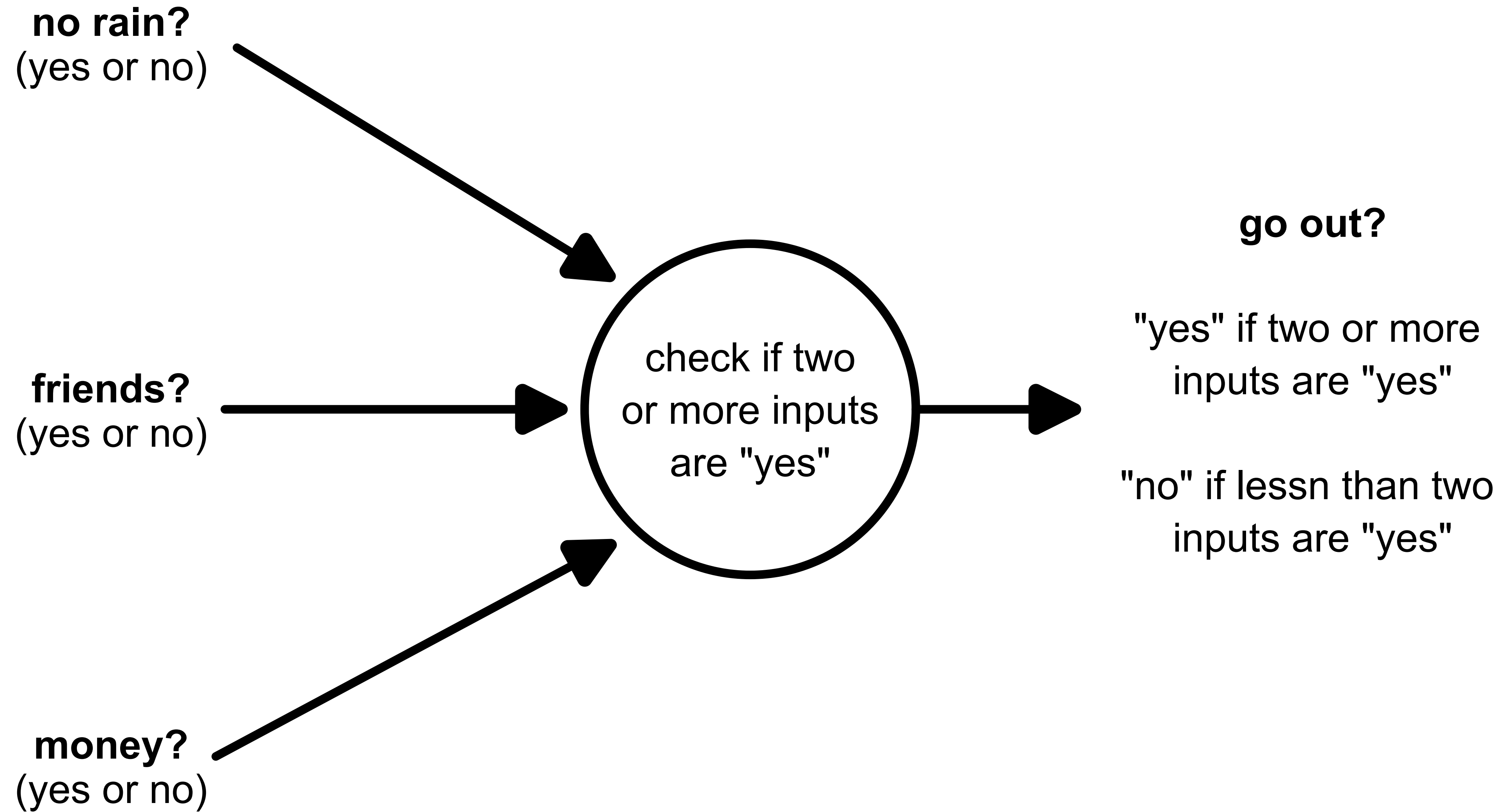


The Perceptron



Should I Go Out Tonight?

- Inputs:
 - Is it not raining outside?
 - Are my friends going out?
 - Do I have money left?
- Output: Go out? (yes or no)
- Simple algorithm: If two or more inputs are "yes", I'm going out.



no rain?
(1 or 0)

1 for "yes"
0 for "no"

friends?
(1 or 0)

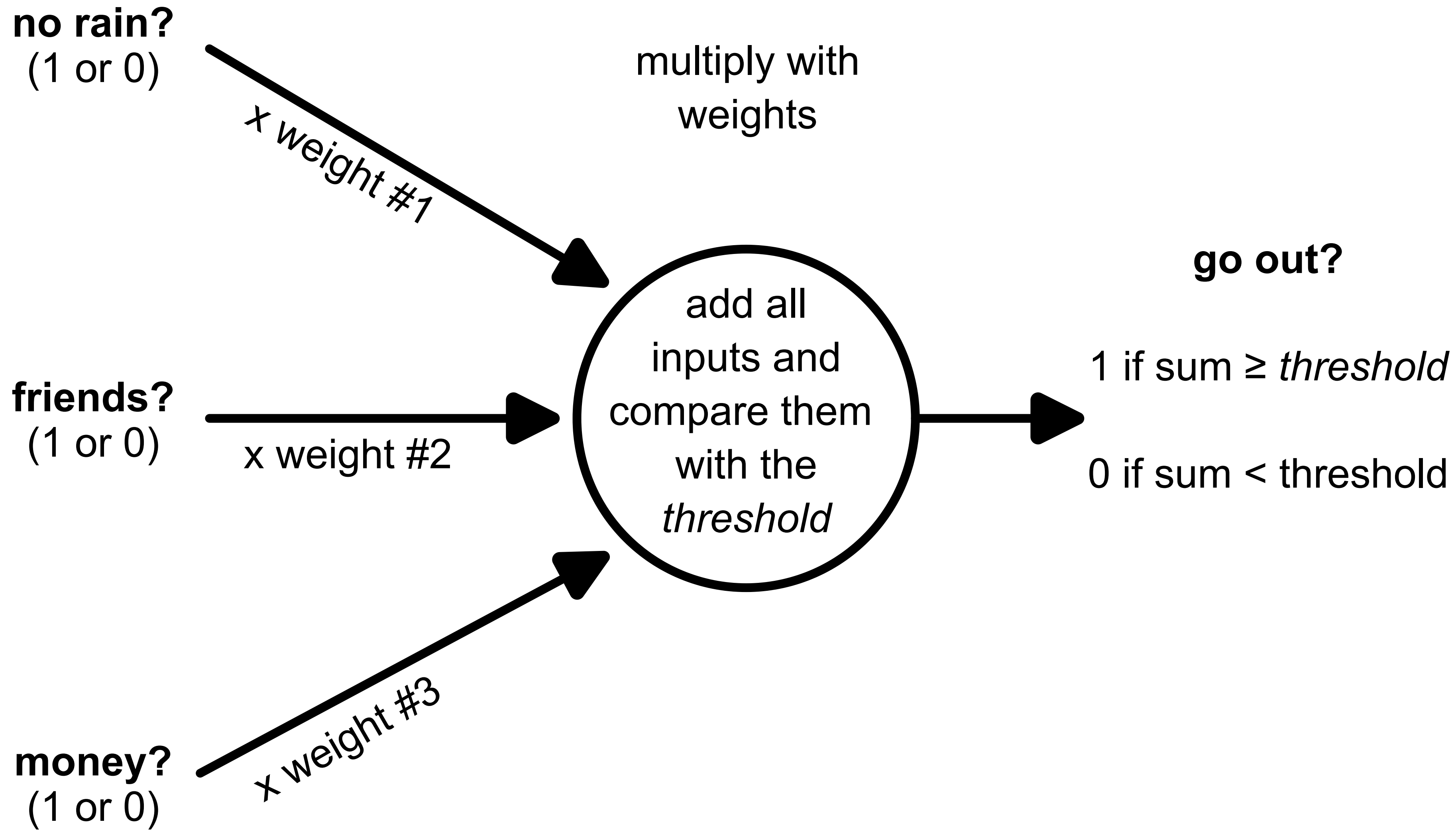
money?
(1 or 0)

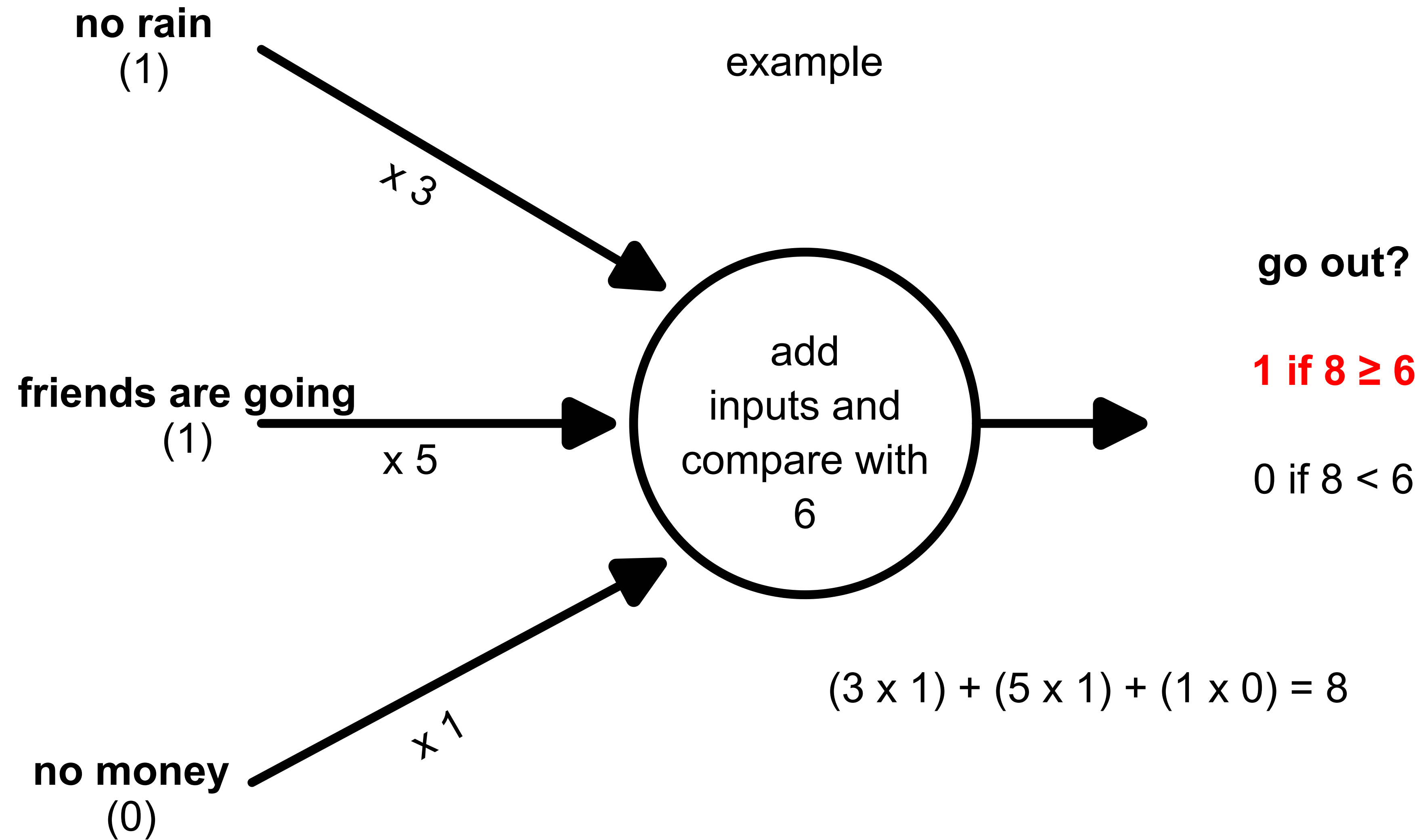
add all
inputs and
compare them
with 2

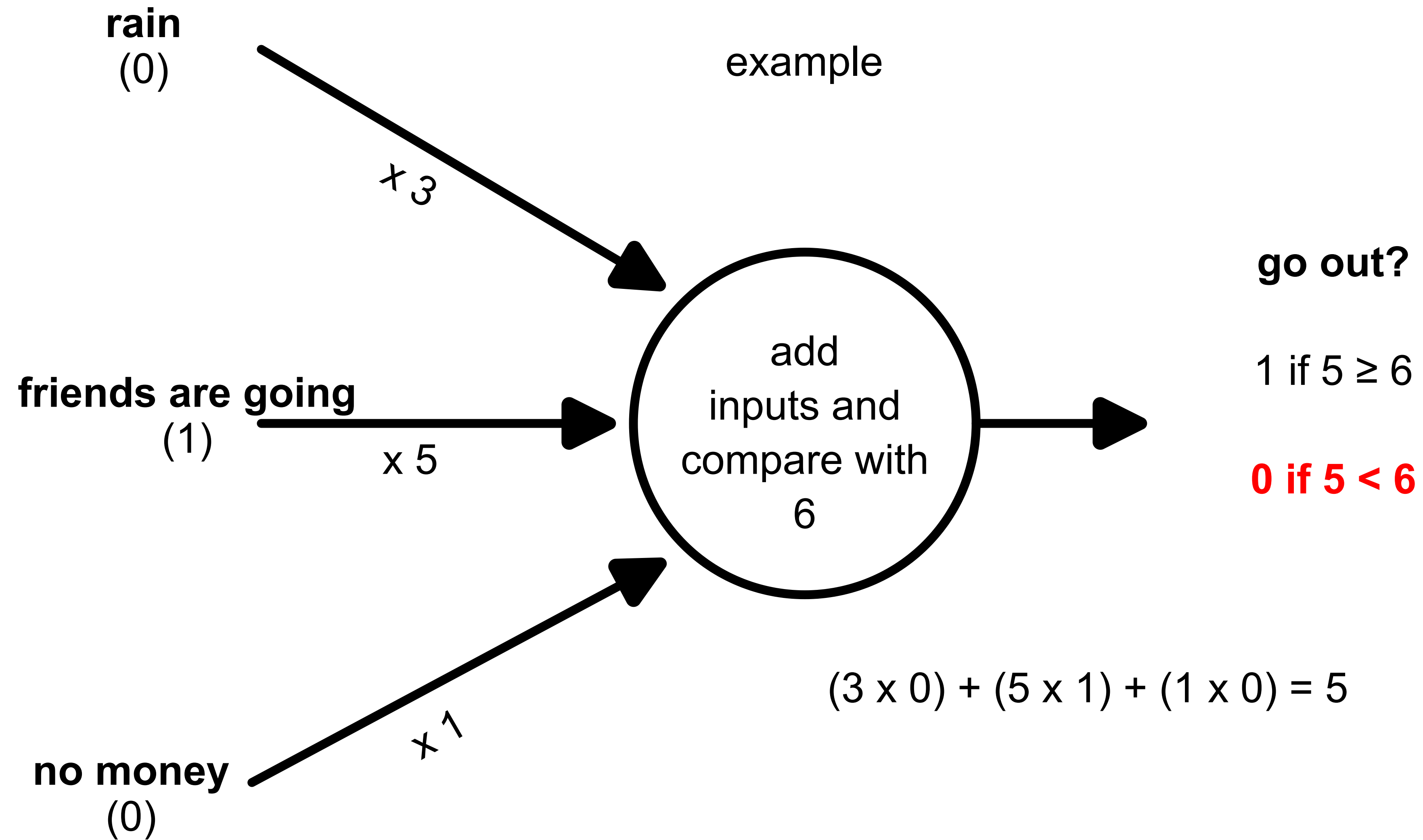
go out?

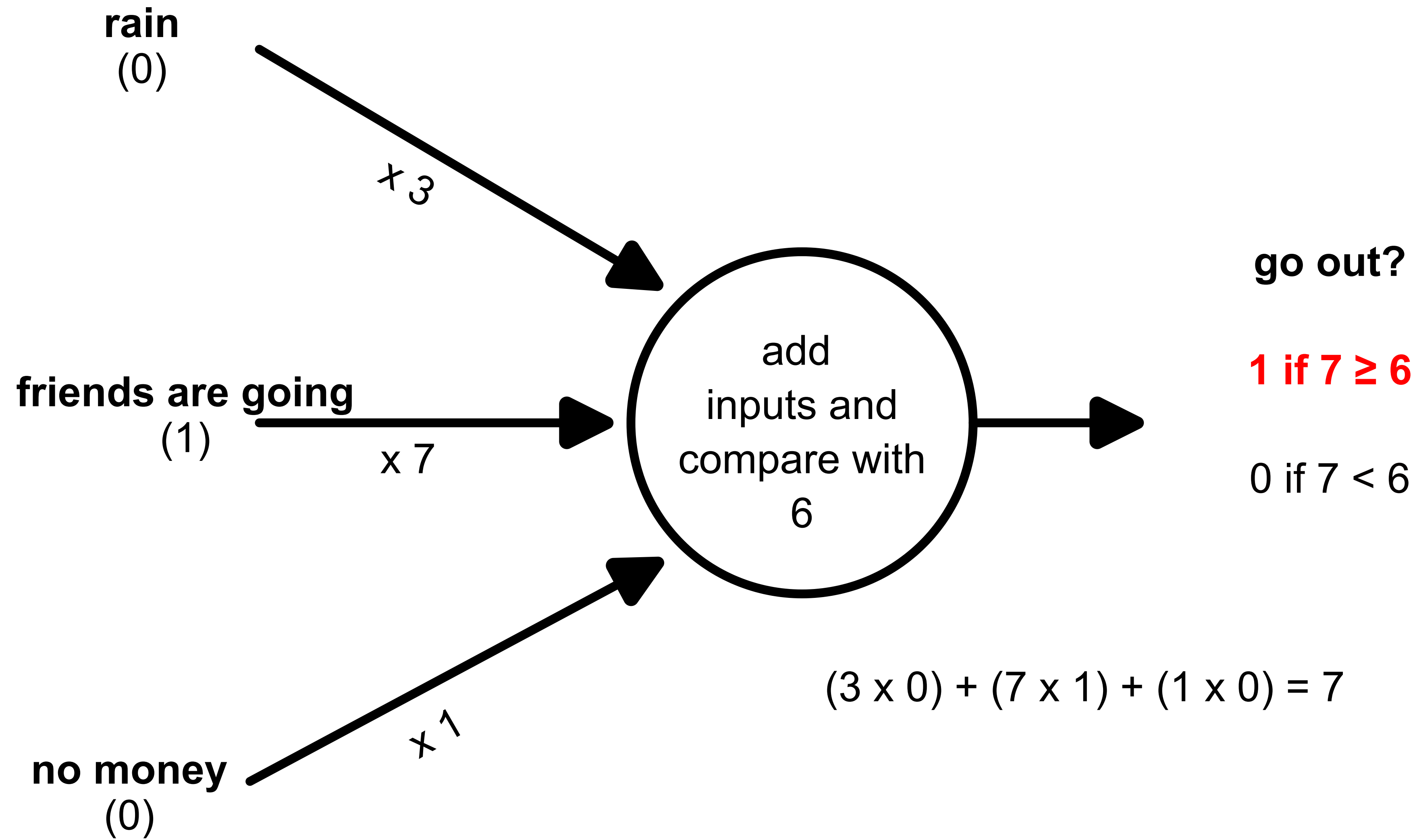
1 if $\text{sum} \geq 2$

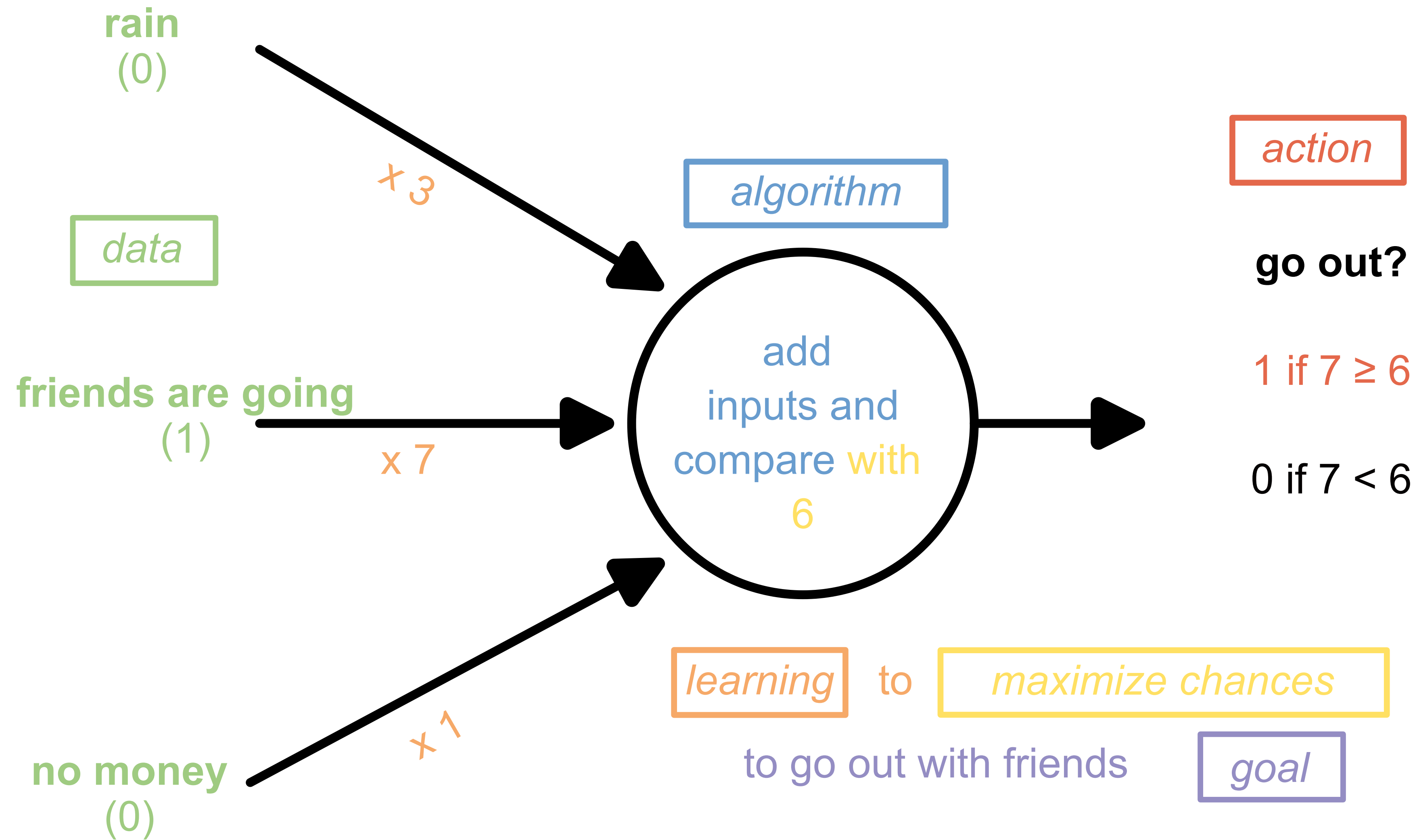
0 if $\text{sum} < 2$



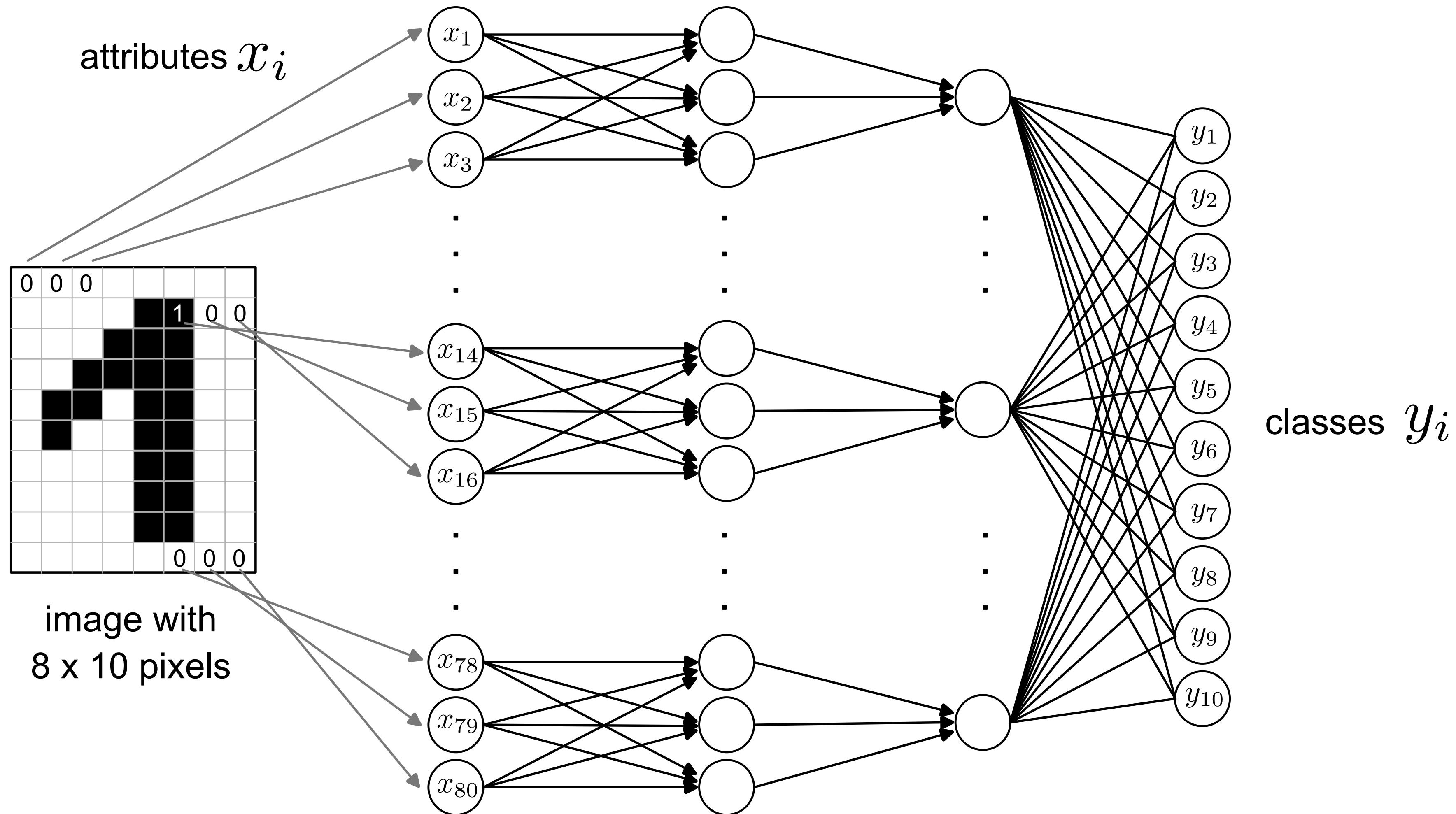




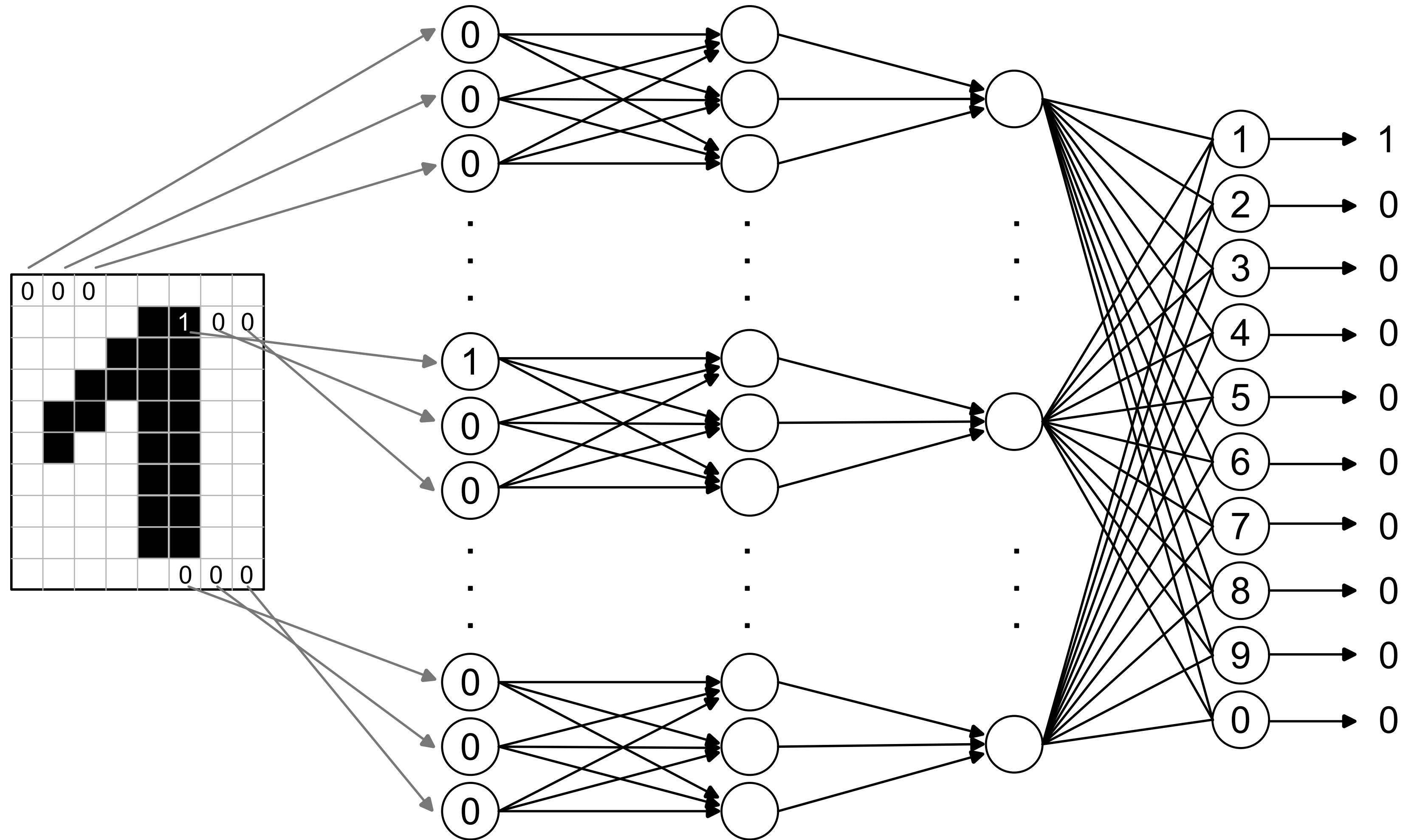




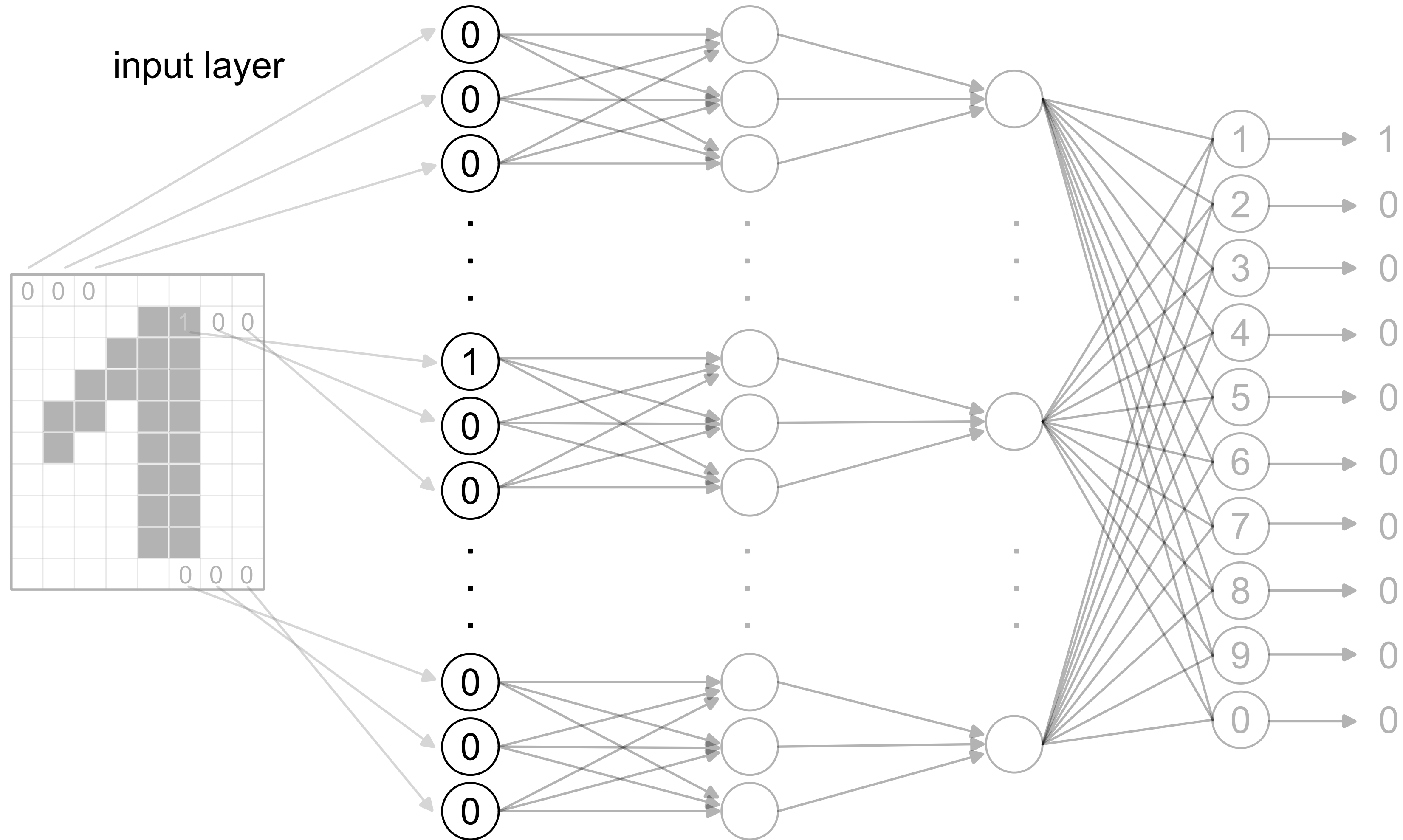
Neural Network for Image Recognition



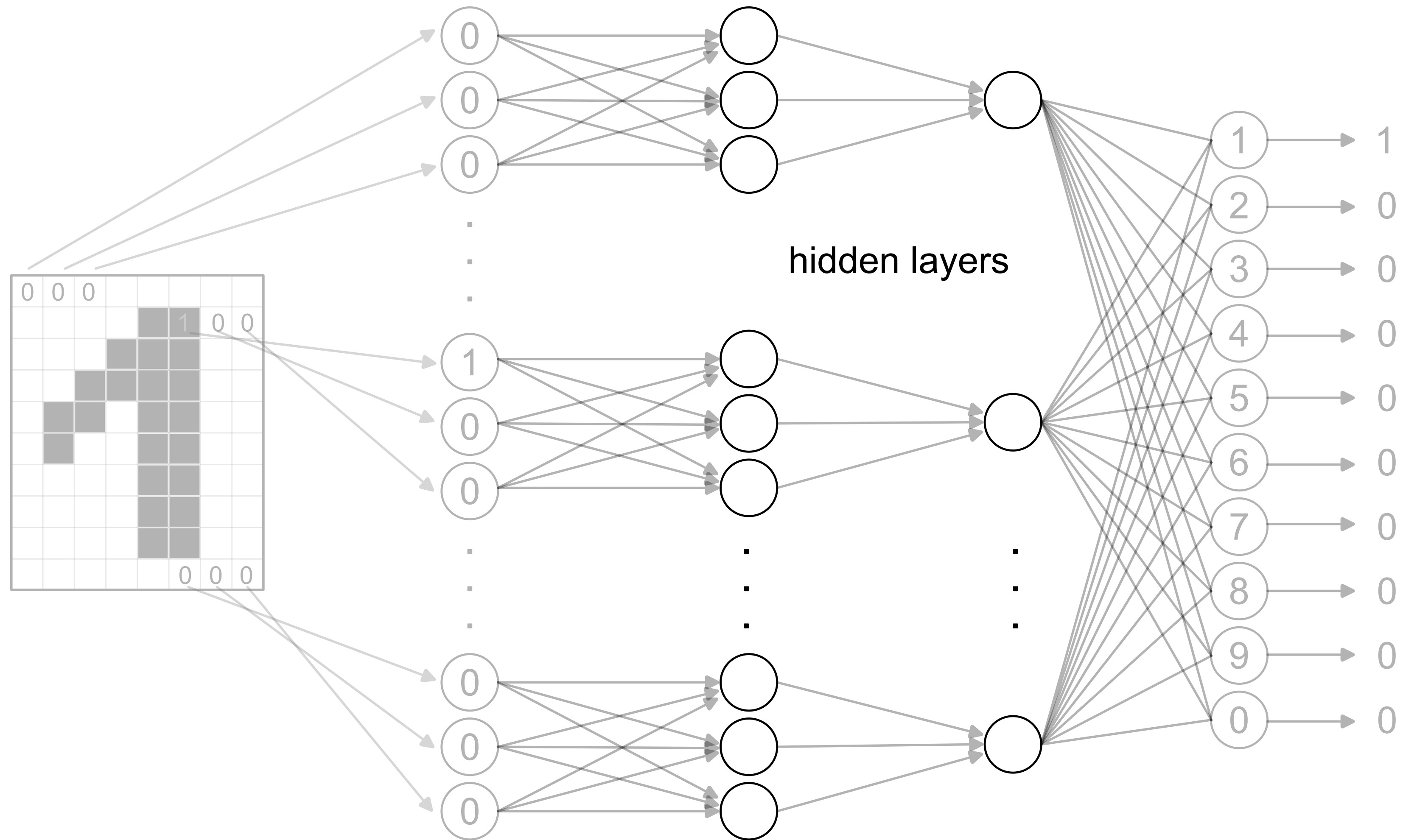
Neural Network for Image Recognition



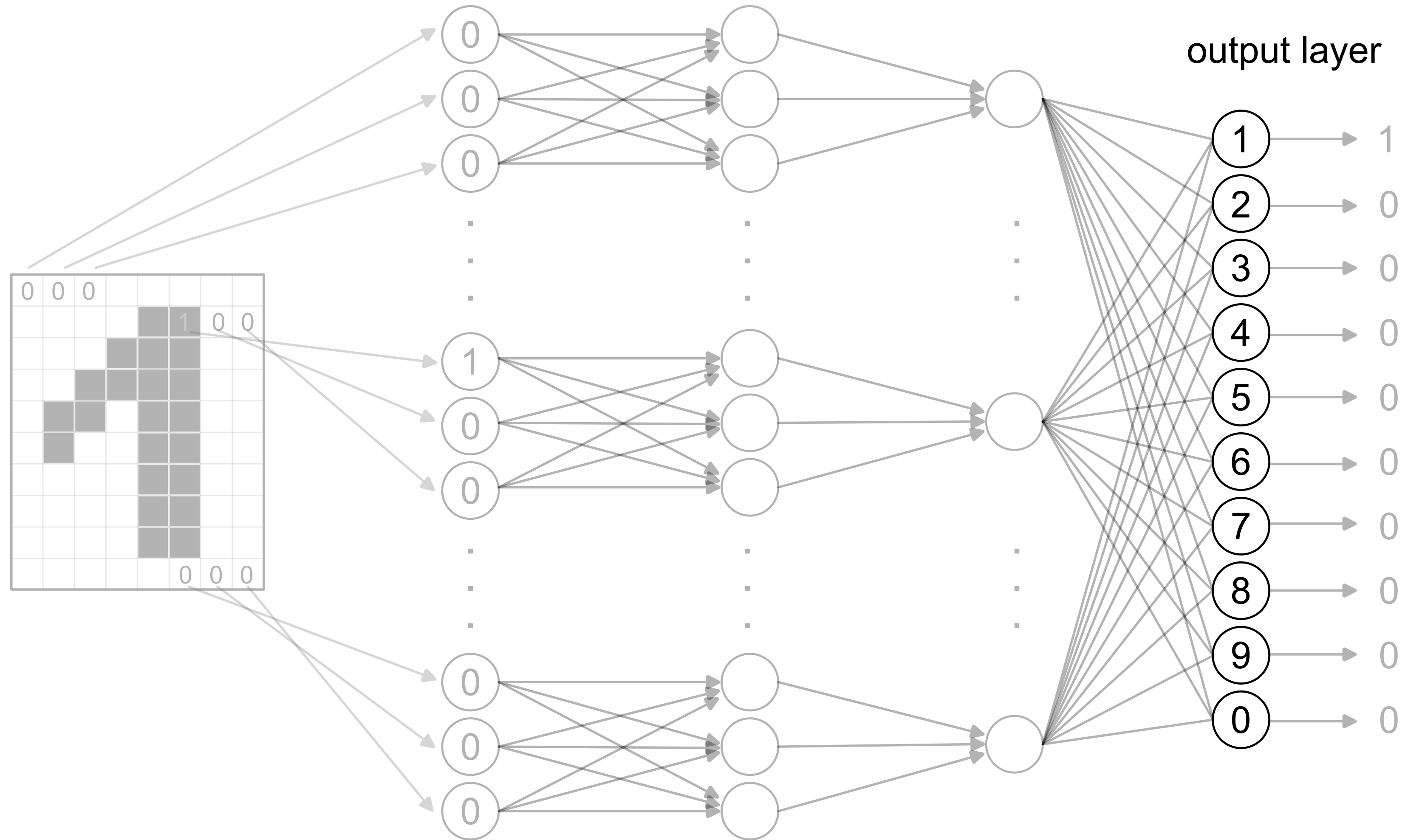
Neural Network for Image Recognition



Neural Network for Image Recognition



Neural Network for Image Recognition



Summary

- Neural networks consist of individual neurons (perceptrons).
- Individual neurons receive **inputs** and calculate an **output** using an **activation function**.
- Outputs are passed to neurons in the next **layer** of the neural network until the output layer is reached.
- The input layer receives the **attributes of an observation** x_i as input. The output layer outputs one of the possible **classes** y_i .
- The importance of inputs can be modified via **weights**.
- During the **training** of neural networks, the weights are adjusted until the output is **optimal**.